Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/16/2006
PI Name:	Abbud-Madrid, Angel Ph.D.		
Project Title:	Fire Suppression and Safety in Reduced Gravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	aabbudma@mines.edu	**Fax:**	FY (303) 384-2327
PI Organization Type:	UNIVERSITY	**Phone:**	(303) 384-2300
Organization Name:	Colorado School of Mines		
PI Address 1:	Center for Space Resources		
PI Address 2:	1310 Maple St.		
PI Web Page:			
City:	Golden	**State:**	CO
Zip Code:	80401	**Congressional District:**	7
Comments:			
Project Type:	INTERDISCIPLINARY (2004-2007)	**Solicitation / Funding Source:**	02-OBPR-08
Start Date:	03/03/2003	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Wu, Ming-shin	**Contact Phone:**	216-433-3781
Contact Email:	Ming-shin.Wu@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NNC04GA97G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This investigation is intended to answer fundamental questions about the suppression mechanisms and the suppression effectiveness of carbon dioxide (CO2) and water-mist on gas-jet and liquid-pool fires in different gravitational environments, such as the ones experienced by spacecraft in orbit and on the surface of the Moon and Mars. The objectives of the proposed investigation are to understand and quantify: 1) the relative importance of various postulated suppression mechanisms, and 2) the relative suppression effectiveness of carbon dioxide (CO2) and water-mist on both gas-jet and liquid-pool fires in different gravitational environments. These objectives will be achieved by conducting experiments at the Colorado School of Mines fire suppression facility to validate the results of numerical simulations conducted at the Naval Research Laboratory (NRL) for specific conditions at normal-gravity and then using the validated simulation tool to predict the effects of reduced gravity on fire suppression in large (rack-scale) compartments. Conducting space-based testing of fire suppression in large compartments is a daunting task from both the safety and the cost points of view. Simulations using a validated and verified numerical tool will provide valuable information that can		

	complement and reduce the number of tests needed, as well as help design better and safer experiments for testing fire-safety policies and practices. Specifically, a computational tool will be developed to characterize the flow field in space compartments and to investigate the transport of fire-fighting agents and their interaction with the air and fuel surrounding a fire. By understanding and quantifying the relative effectiveness of CO2 and water-mist for fire suppression, a more informed decision can be made on the choice of fire suppressants for future spacecraft.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will also provide additional basic information on fire suppression and thus will enhance our knowledge base for fighting fires on Earth, specially fires on enclosed spaces such as aircraft cabins and cargo bays, ships, machinery and telecommunication rooms, museums, and libraries.
Task Progress:	The investigation during the 2005 fiscal year focused on finishing the development of the apparatus for the methane gas jet burner suppression experiments and performing the first set of tests to gather information on flame heights, fuel flow, and fuel velocity with gas-jet flames with no suppression agent present. The initial experiments consisted of flowing methane through a 1.25-cm diameter stainless steel tube at various flow rates varying from 0 to 300 sccm, corresponding to flow velocities of approximately 0-240 cm/min. It was noticed that the flame height grows linearly with flow velocity up to an approximate velocity of 240 cm/min, after which the flames become unstable and tend to flicker. Subsequent tests included an incursion into the unstable flow region to replicate flame conditions encountered in turbulent gas-jet flames, which correspond to typical flames found in most fire scenarios. Video images and temperature measurements were taken from flames with methane fuel flows up to 1000 sccm ejecting into open air in the combustion chamber. For the unstable flow tests, the images and temperature data show two buoyancy driven flame fluctuations consisting of a rapid fluctuation at a frequency of approximately 5 Hz, which is at the same time embedded in another flame fluctuation with a smaller frequency of approximately 1 Hz. These preliminary data were provided to the Naval Research Laboratory (NRL) modeling team for calibration and verification of their numerical model. Once their numerical validation is complete, we will provide them with data from the suppression experiments which include flame temperature, video images, and water/gas concentration during the flame suppression event with water mist and carbon dioxide, as well as the extinction time and behavior observed.
Bibliography Type:	Description: (Last Updated: 03/21/2008)
NASA Technical Documents	Abbud-Madrid A, McKinnon JT, Delplanque JP, Kailasanath K, Gokoglu S, Wu M. "Spacecraft Fire Suppression: Testing & Evaluation." In: Workshop for Strategic Research to Enable NASA's Exploration Missions, NASA/CP-2004-213205, Vol. II, pp. 1- 11. Cleveland, Ohio : NASA John H. Glenn Research Center at Lewis Field, 2004. , Jun-2004
Presentation	Schwer, D. A.; Kailasanath, K., and Abbud-Madrid, A. "Fire Suppression and Safety in Reduced Gravity Environments" Microgravity Fire Modeling Working Group Meeting, NASA John H. Glenn Research Center at Lewis Field, Cleveland, Ohio Apr-2005
Presentation	Schwer, D. A.; Kailasanath, K., and Abbud-Madrid, A. "Development of a Water-Mist Fire Suppression Tool for Low-Gravity Fire Suppression Modeling " Fall Technical Meeting of the Eastern States Section of the Combustion Institute, Orlando, Florida Nov-2005